

TERM PROJECT– CASE STUDY

MATH 60611A

1. General instructions

- This assignment counts for 35% of the final grade
- There are three parts, with due dates:
 - o March 10th, 2025: part 1 (report)
 - o April 20th, 2025: part 2 (slides)
 - o April 20-25th: oral interview (date to be defined with the professor)
- Submissions must be made through Zone Cours
- To be done in teams of 2 students.
- The data you will use for this project is in the *data_final.csv* file. A dictionary of file variables is also attached.

2. Pedagogical goals

The MATH 60611A course is an advanced course. At this point in your studies, you should have the tools to solve real problems and come up with innovative solutions in a non-academic context. You should also be able to learn by yourself new concepts and methods. The purpose of this project is to put you in such a situation, and to expose you to a case whose problem is defined in a general way by decision-makers who do not have specific knowledge of data science. This project will also give you the opportunity to work with spatial data, something you probably haven't encountered in any of your courses at HEC.

3. Description of the case

You are a consultant hired by the City of Montreal as part of a road safety pilot project. Faced with the constant increase in the number of accidents involving pedestrians on its road network, the city decided to carry out a study to assess the level of risk at road intersections. To do this, a sample of 1864 signalized intersections was selected (see figure 1). This sample represents 80% of all signalized intersections on the island of Montreal. For each intersection, the city has the total number of accidents involving pedestrians and reported by the police (data from the SAAQ) in the last 10 years. In addition, Department of Transportation engineers collected the following data:

- Traffic data on the average daily flows of pedestrians and vehicles at each intersection
- Geometry data (design) for each intersection including the number of lanes, types of signals, layout for left turns, parking restrictions, etc...
- Road safety performance measures calculated by engineers for each intersection, based on the number of potential daily interactions between pedestrians and vehicles, depending on the directions and movements of the vehicles.

The objective of your mandate is to provide the city with a ranking of the 1864 intersections in terms of safety (from the most dangerous to the least dangerous), so that it can prioritize the riskiest intersections with the aim of improving infrastructure.

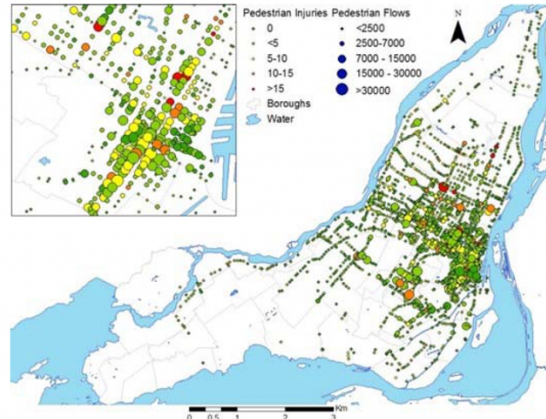


Figure 1 Signalized Intersections with Pedestrian Injuries and Flows

4. General objective

Your analysis should be based on a modeling of the accidents observed in the last 10 years (variable acc). You are completely free to choose the methods of your choice, there is no single answer to this problem and several solutions are adequate. Demonstrate creativity, depth in your analyses, and rigor in the implementation. The data you receive is reliable, but it is up to you to determine which relevant information you want to use.

5. Specific objectives and deliverables

PART 1 [50%]

Answer the research question using the methods covered in class (variable selection, non-linear regression models, tree-based methods, boosting), while ignoring the spatial correlation present in the data.

The deliverables for this part are:

- A report of no more than 8 pages (excluding code), clearly describing the relevant descriptive analysis, data pre-processing steps and analytical approach
- A file containing your cleaned R code to reproduce the results presented.
- A .csv file containing 2 columns: the ID of the intersection and its rank in terms of risk (a rank of 1 represents the most dangerous intersection).
- Three slides (ppt) summarizing your main findings for the city, tailored to a non-technical audience

PART 2 [50%]

Answer the research question using some methods specifically designed for spatial data, or methods that account for the spatial correlation between intersections. For this part, you will

need to read and explore the literature on spatial data analysis and come up with an innovative solution. Note that there are many ways one can approach this problem, I expect that each of you will provide a unique solution.

The deliverables for this part are:

- A report explaining your analysis strategy, the methods used, and the results in **slide format** (ppt or Latex). These slides are designed to be read by the professor, meaning they should contain enough detail to be understood without a formal presentation (you can refer to the pdf course notes provided by the professor as an example). The report should be a maximum of 30 slides. While there are no specific guidelines for font size, the slides should be prepared as if they were going to be projected on a screen for a large audience.
- A file containing your cleaned R code to reproduce the results presented.
- A .csv file containing 2 columns: the ID of the intersection and its rank in terms of risk (a rank of 1 represents the most dangerous intersection).

6. Assessment

Part 1 will be evaluated based on the following criteria:

- Depth of your analyzes
- Accuracy of your general approach
- Appropriate use of methods
- Clarity of the report
- Relevance of your analyzes for the city of Montreal
- Clarity and relevance of the results presented to the city (ppt slides)
- Reproducibility of your results: your code must be able to compile without errors and generate the ranks presented in the deliverables.

Part 2 will be evaluated based on the slides provided **and an oral interview** using the following criteria:

- Understanding of the methods used
- Depth of your analyzes
- Appropriate use of methods
- Clarity of the slides
- Reproducibility of your results: your code must be able to compile without errors and generate the ranks presented in the deliverables.

Note1: the oral interview will be conducted by teams, but the grade given for Part 2 will be given individually

Note2: you need to register for the oral interview date (first come, first served). The instructions for registration will be sent by the professor.